

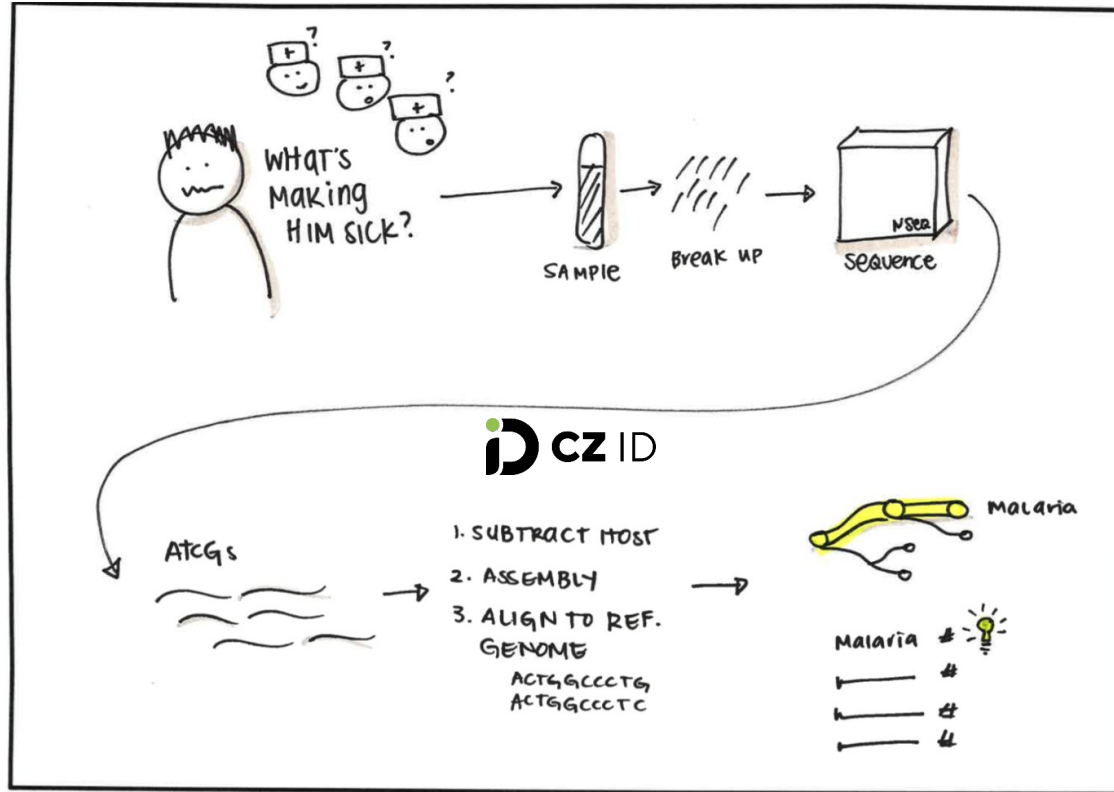
Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



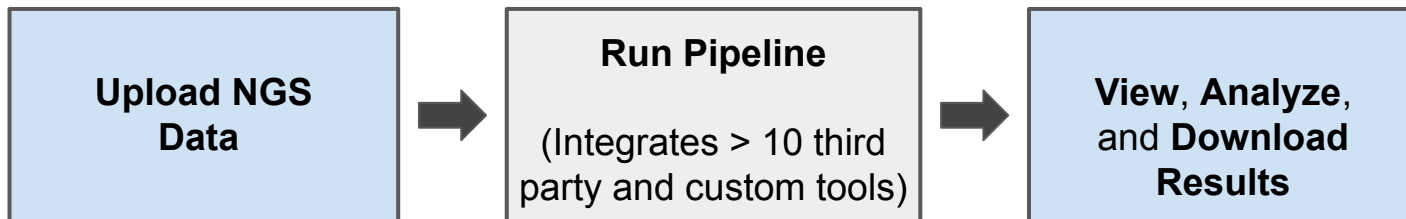
Introduction to CZ ID

CZ ID is a free platform
used to identify microbes of interest from NGS data



CZ ID integrates bioinformatic tools to enable NGS analysis regardless of available computational resources

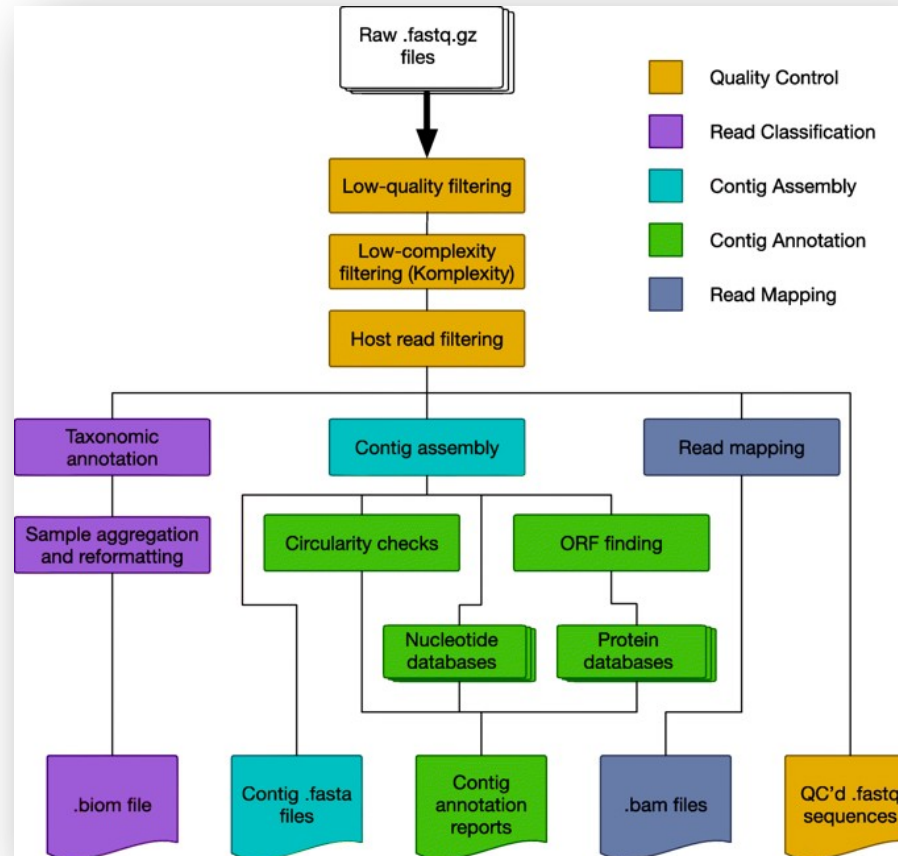
General Workflow:



Plug-and-play, cloud-based platform

(optimized for scalable Amazon Web Services (AWS) cloud deployment)

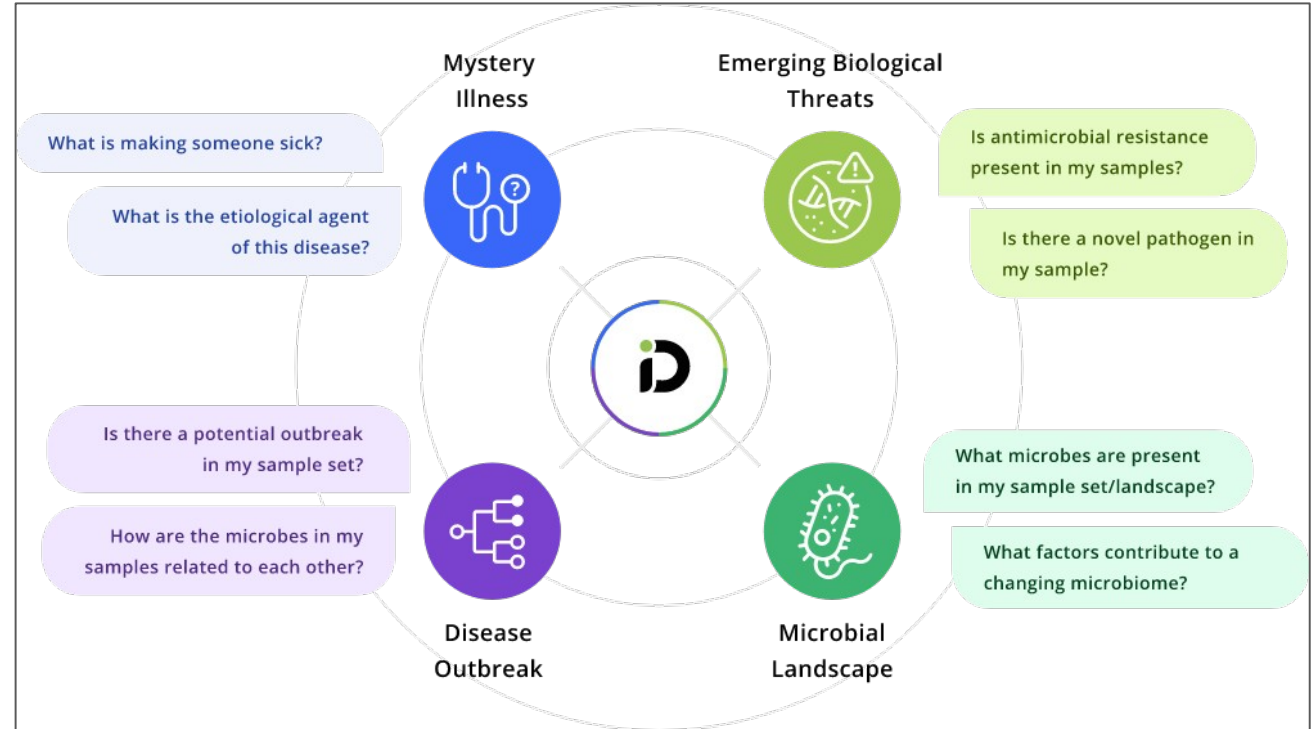
NGS data analysis requires computational power and includes MANY steps



CZ ID supports multiple uses

CZ ID Pipelines:

- Metagenomic (mNGS) analysis
- Antimicrobial Resistance (AMR) gene detection
- Viral consensus genome assembly



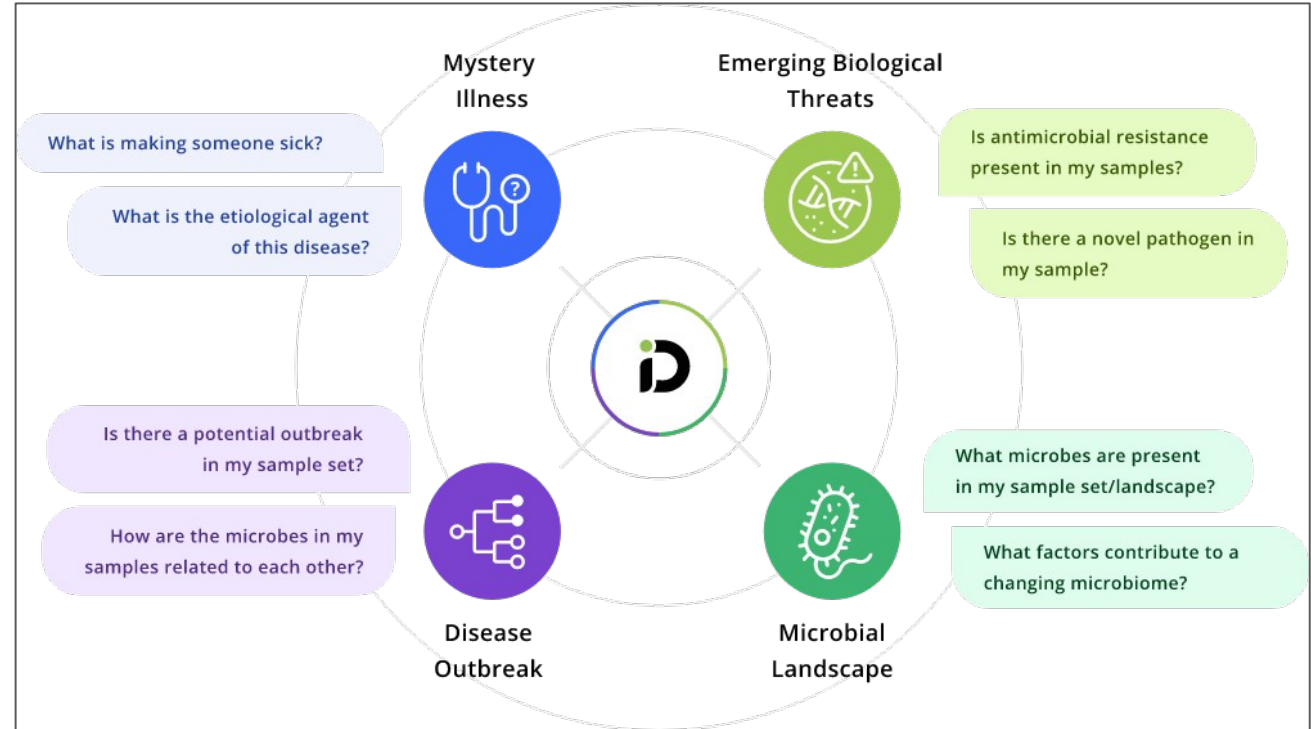
CZ ID supports multiple uses

Module	Input Sequencing Data	Supported Data
Metagenomics (mNGS)	Shotgun (or random) data Note: Module is NOT meant to process amplicon data (e.g., 16S, 18S, ITS)	Illumina & Nanopore
Antimicrobial Resistance (AMR)	Shotgun (or random) and whole genome sequence data	Illumina
SARS-CoV-2	Metagenomic sequencing with spiked primer enrichment (MSSPE) and amplicon-based data (e.g., ARTIC v3 protocol)	Illumina & Nanopore
Viral Consensus Genomes	Target enrichment (e.g., MSSPE), PCR amplicon, whole genome, or metagenomic sequencing data	Illumina

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CZ ID Pipelines:

- **Metagenomic (mNGS) analysis**
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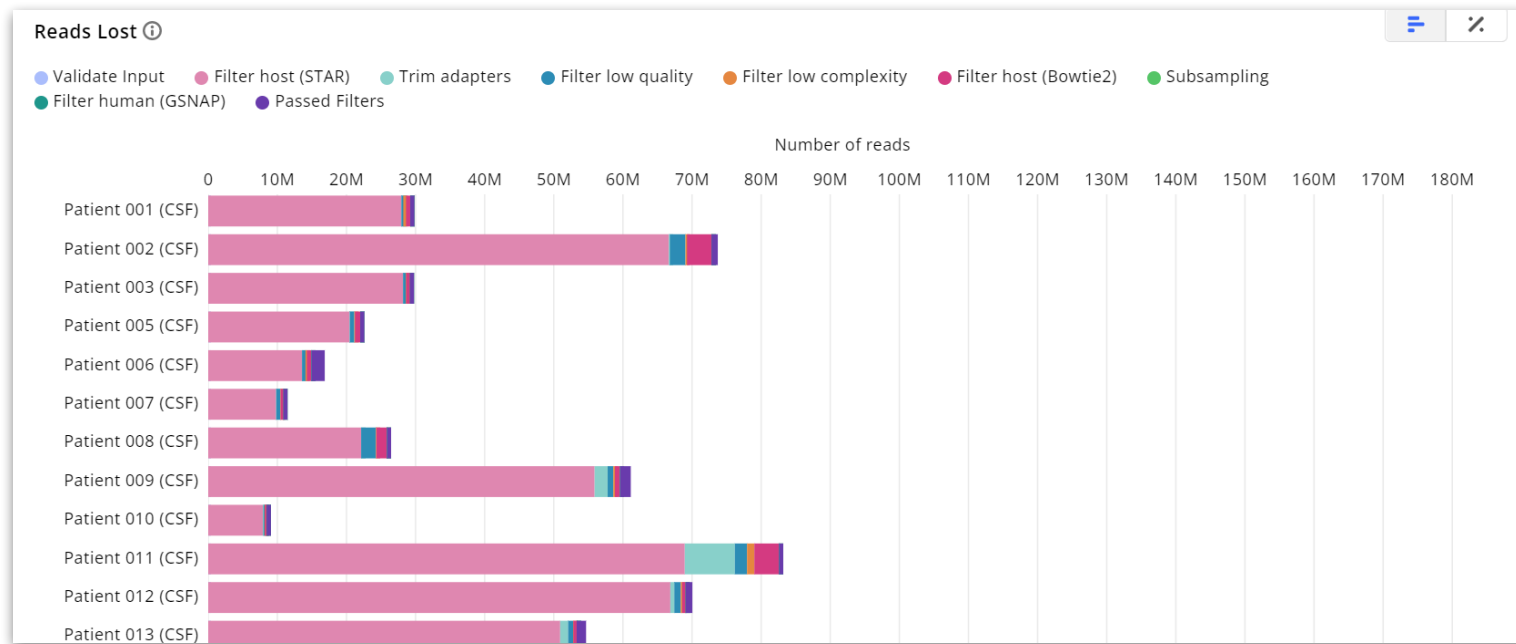


CZ ID mNGS Analysis Highlights

With just a few clicks, easily:

- Evaluate sample quality
- Explore and evaluate identified taxa through interactive reports
- Compare samples using interactive heatmaps
- Generate viral consensus genomes
- Keep track of pipeline steps

Perform QC



Explore and evaluate metrics for detected microbes

Medical Detectives

Illumina mNGS Pipeline v6.1 | processed 3 years ago | SAMPLE DETAILS

Patient 010 (CSF)

Share

Download

Metagenomic

Antimicrobial Resistance (Deprecated)

Consensus Genome

Taxon name

Name Type: Scientific

Background: MedicalDetectives_WaterCtrls

Categories

Threshold filters: 1

Read Specificity: Specific Only

Annotation

NT rPM >= 10

112 rows passing the above filters, out of 3100 total rows

CLEAR FILTERS

	Taxon	Score	Z ...	rPM	r	con...	con...	%...	L	E v...	NT NR
▼	Alphavirus (1 viral species: 1)	72,976,252	100.0 100.0	3,650.9 3,646.7	19,344 19,322	1 1	19,139 19,139	100.0 99.9	11,671.6 2,450.2	10 ⁻³⁰⁵ 10 ⁰	
	Chikungunya virus Known Pathogen	72,976,252	100.0 100.0	3,650.9 3,646.7	19,344 19,322	1 1	19,139 19,139	100.0 99.9	11,671.6 2,450.2	10 ⁻³⁰⁵ 10 ⁰	
>	Tetraparvovirus (1 viral species:)	46,706,236	100.0 100.0	2,366.2 2,304.6	12,537 12,211	2 1	12,102 11,987	99.1 99.6	4,677.3 896.3	10 ⁻²⁹⁶ 10 ⁰	
>	Providencia (1 bacterial species: 1)	14,332,531	100.0 0.0	1,433.3 0.0	7,594 0	1 0	7,594 0	99.8 0.0	1,354.0 0.0	10 ⁻³⁰⁸ 0	
>	Lactobacillus (1 bacterial species:)	2,502,625	100.0 100.0	28.9 293.5	153 1,555	7 9	79 1,475	98.6 78.8	209.0 157.7	10 ⁻¹⁰⁸ 10 ⁻¹	
>	Rothia (1 bacterial species:)	2,364,849	100.0 100.0	238.4 5.5	1,263 29	6 0	1,127 0	96.5 97.3	2,086.8 45.7	10 ⁻²⁷² 10 ⁻¹⁶	

Example

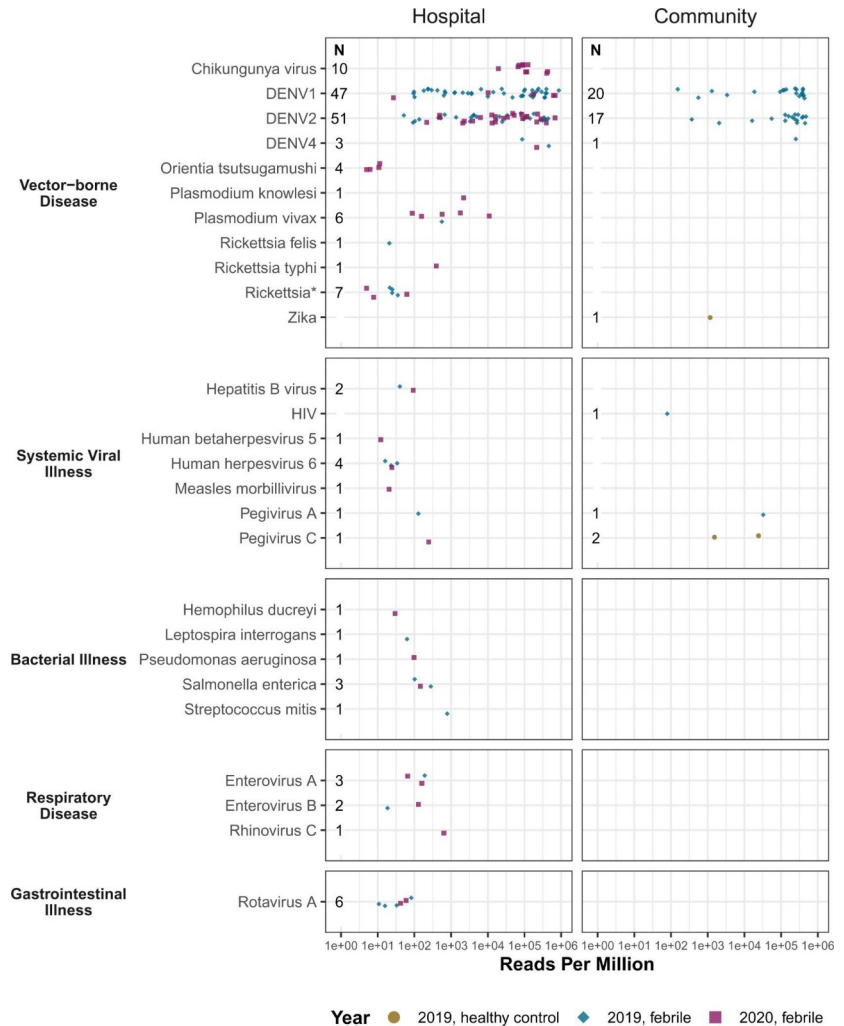
Study: Pathogen landscape associated with febrile illness in Cambodia

Sample type: Serum

Results: 203 pathogens detected from 489 samples

Figure: Pathogens organized by disease category. Detection distinguished by participant cohort (community- (n = 87) vs hospital-based cohorts (n = 377))

[Bohl et al 2021](#)



Example

Study: Pathogen landscape associated with pediatric febrile illness in Tororo, Uganda

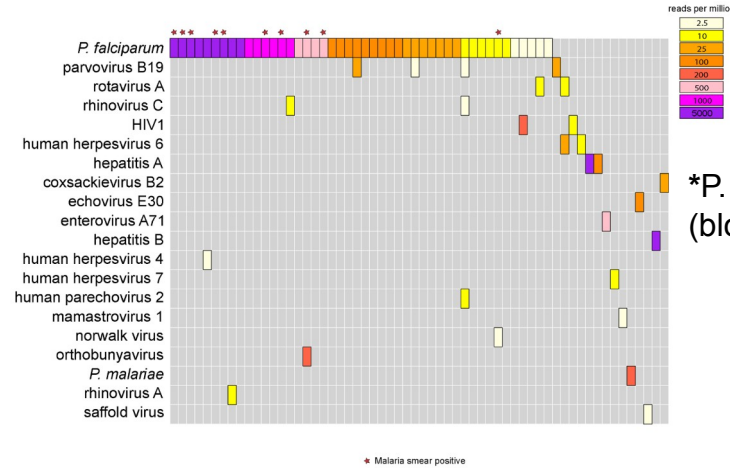
Sample type: Serum and Nasopharyngeal (NP) swabs

Results: Pathogens detected in more than half of serum and NP samples.

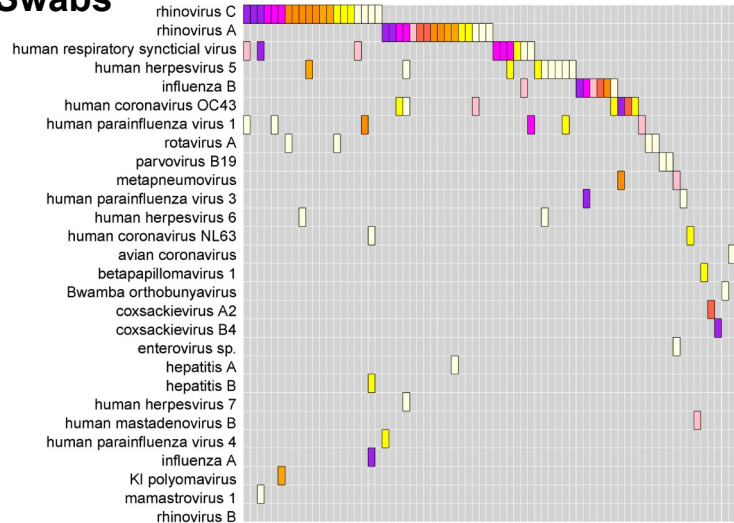
Figure: Pathogens organized by sample type

[Ramesh et al 2019](#)

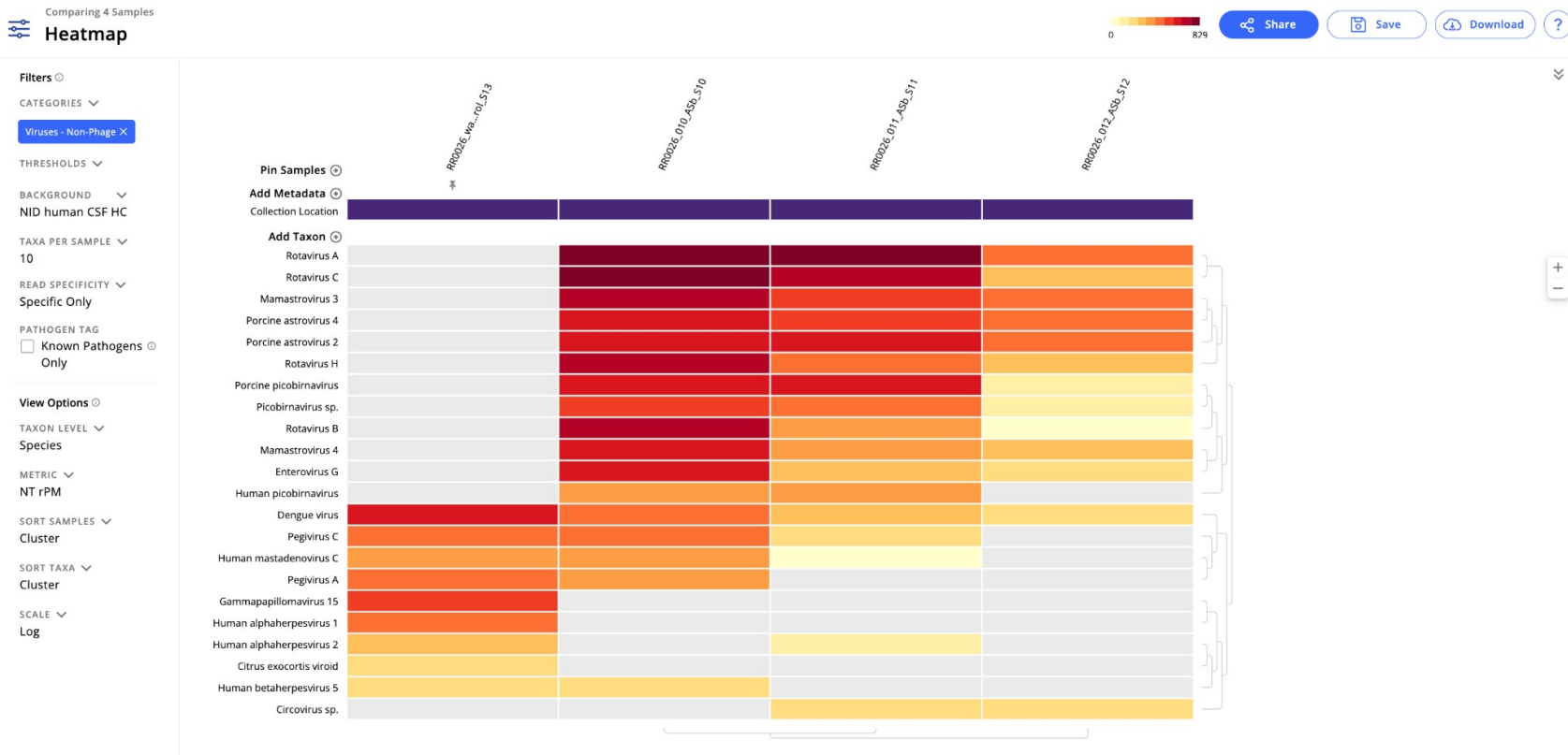
Serum



NP Swabs



Compare samples visually to identify patterns of interest



Create viral consensus genomes

Is my consensus genome complete? ⓘ

Taxon	Mapped Reads	GC Content	SNPs	%id	Informative Nucleotides	% Genome Called	Missing Bases	Ambiguous Bases
Chikungunya virus	1048870	50.1%	69	99.4%	11713	100%	1	0

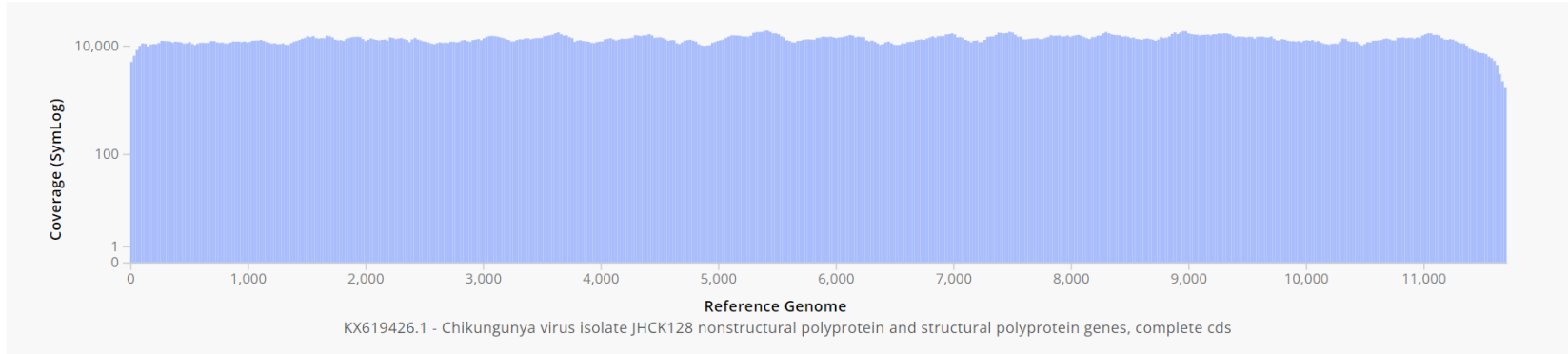
How good is the coverage? ⓘ

Reference NCBI Entry [KX619426.1](#)

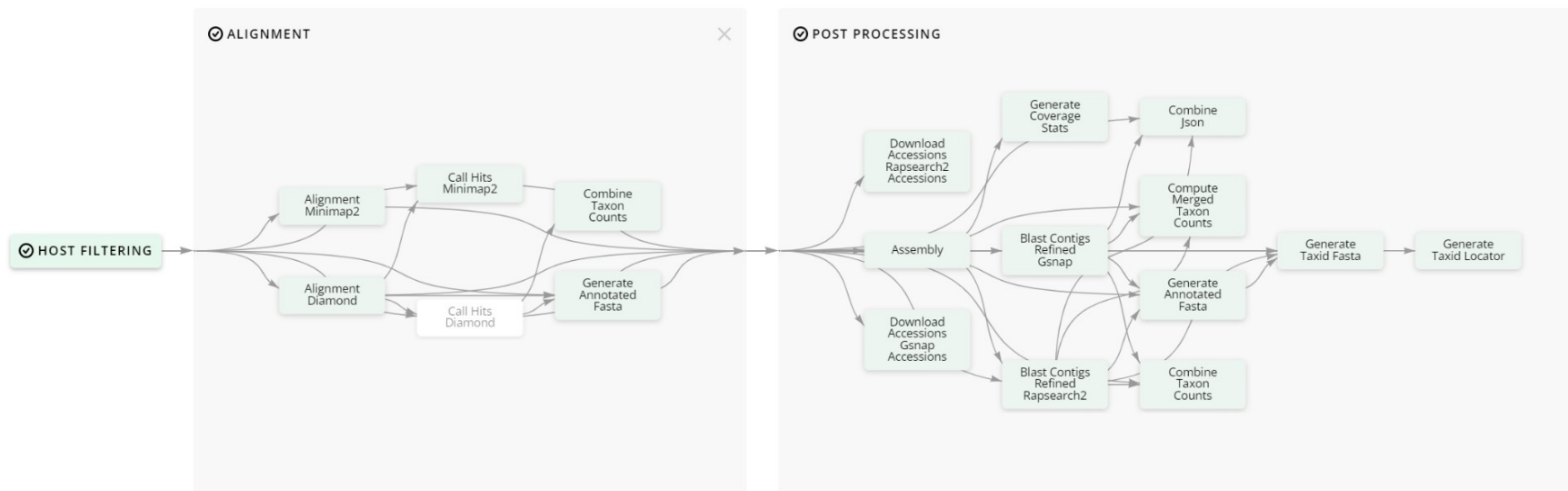
Reference Length 11708

Coverage Depth 13075.0x

Coverage Breadth 100.0%



View pipeline steps and access intermediate files



Pipeline visualization

Publications Describing CZ ID Pipelines

- CZ ID (formerly IDSeq) platform and mNGS Pipeline
 - [IDseq—An open source cloud-based pipeline and analysis service for metagenomic pathogen detection and monitoring](#)
- CZ ID AMR Pipeline
 - [Simultaneous detection of pathogens and antimicrobial resistance genes with the open source, cloud-based, CZ ID pipeline](#) (preprint)
- CZ ID Nanopore mNGS Pipeline
 - [CZ ID: a cloud-based, no-code platform enabling advanced long read metagenomic analysis](#) (preprint)

Example Publications

- [Pathogen genomics in public health](#)
- [Unbiased metagenomic sequencing for pediatric meningitis in Bangladesh reveals neuroinvasive Chikungunya virus outbreak and other unrealized pathogens](#)
- [Integrating central nervous system metagenomics and host response for diagnosis of tuberculosis meningitis and its mimics](#)
- [Metagenomic sequencing of post-mortem tissue samples for the identification of pathogens associated with neonatal deaths](#)
- [Discovering disease-causing pathogens in resource-scarce Southeast Asia using a global metagenomic pathogen monitoring system](#)

And more!

Let's look at the pipeline...

But first, for ONT only: DL3-O_Prepare ONT fastqs for upload to CZ ID

Next Slides, Illumina: DL4-I_CZ ID mNGS Pipeline for Illumina

Next Slides, ONT: DL4-O_CZ ID mNGS Pipeline for ONT